

1

Impact of Civil Engineering : An Introduction

UNIT SPECIFICS

Through this unit we have discussed the following aspects:

- ***The past to look into the future:***
 - *Pre- industrial revolution*
 - *Agricultural revolution*
 - *first and second industrial revolutions*
 - *IT revolution*
- ***Major Civil Engineering breakthroughs and innovations - a Timeline***
- ***Sustainable Development - Present day world and future projections***
 - *The steady erosion in Sustainability and its global impact*
 - *Sustainable Development Goals (SDGs), its impact and possible causes*
- ***Evaluating future requirements for various resources***
 - *GIS and applications for monitoring systems*
 - *Human Development Index and Ecological Footprint*

Besides giving a large number of multiple-choice questions as well as questions of short and long answer types marked in two categories following lower and higher order of Bloom's taxonomy, a list of references and suggested readings are given in the unit so that one can go through them for practice.

There is a "Know More" section, which has been carefully designed so that the supplementary information provided in this part becomes beneficial for the users of the book. It is important to note that for getting more information on various topics of interest some QR codes have been provided which can be scanned for relevant supportive knowledge. This section mainly highlights applications of the subject matter for our day-to-day real life or/and industrial applications on variety of aspects, case study related to environmental, sustainability, social and ethical issues whichever applicable, and finally inquisitiveness and curiosity topics of the unit.

Unit 1 - Impact of Civil Engineering: An introduction

RATIONALE

This introductory unit on the historic and socio-economic drivers behind the emergence of Civil engineering and its profound impact on sustainable development sets stage for the chapters ahead.

UNIT OUTCOMES

List of outcomes of this unit is as follows:

U1-O1: Understanding/ comprehension of world history and socio-economic context

U1-O2: Knowledge on the major breakthroughs and innovations of the domain

U1-O3: Knowledge on Sustainable development, its Goals, Impact and Indicators

Unit-1 Outcomes	EXPECTED MAPPING WITH COURSE OUTCOMES						
	(1- Weak Correlation; 2- Medium correlation; 3- Strong Correlation)						
	CO-1	CO-2	CO-3	CO-4	CO-5	CO-6	CO-7
U1-O1	2	2	3	1	2	1	
U1-O2	3	1	2		2	1	
U1-O3	3	2	3		2	3	2

The distinct most character of a civilisation is the advent of organised social and physical infrastructures to improve the quality of civic life. Though formal recognition of this branch and degree first appeared in the early 18th century, the practice and profession of civil engineering has existed since the dawn of civilisation. It is the branch of engineering or professional discipline of; planning, designing, building/construction and maintenance of physical infrastructures and built environment, to serve the general public or civilians. Since the designed built environment is a testament to the cultural heritage of a civilization, and a reflection of the socio-cultural, scientific, and technological advancements of the time, Civil engineering plays a pivotal role in impacting the society and global population.

In the following Section, the historical events that transformed society and challenged the discipline of civil engineering to develop is discussed.

1.1 DAWN OF CIVIL ENGINEERING: A BRIEF HISTORY

1.1.1 Pre-Industrial Revolution

Early human history can be dated back to the Stone Age in the Palaeolithic period (2.5 million years ago – till 10,000 BCE) when early man followed nomadic ways of the hunter-gatherer, lived in caves and huts or *tepees* and had begun to develop rudimentary tools of stone and wood. But around 10,000 BCE, a shift in climate and discovery of farming led the Neolithic man to settle, mostly along riverbanks, with focus on production of food. This in turn, led to the need to build secure habitation, plan settlements, and land use, construct transport and irrigation systems, and design solutions towards water supply and sanitation, which can be identified as the first examples of civil engineering.

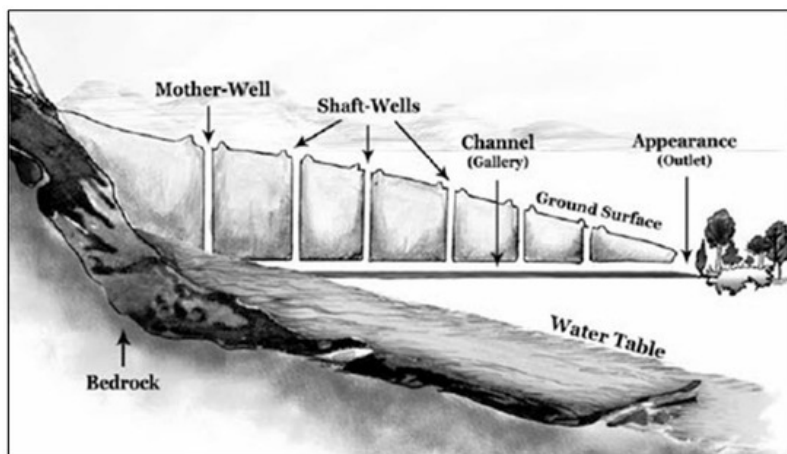


Fig. 1.1 Persian Qanat Profile (source: Manian, et al., 2022)

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In the following era, known as Bronze age, the social organisation altered with kings and leaders emerging and early towns being established, leading to the migration of people for better wages and employments. Skills, tools, and techniques that were handed down from generation to generation of skilled artisans, stonemasons, and carpenters, in a small locale developed into expertise and artistry that led to the rise of planners, architects and engineers, tackling projects traversing regions and empires. Between 4000 and 2000 BC, the profession grew from being utilitarian to being a creative practice of achieving feats of grandeur as is exemplified in the intricate planning of urban housing, sanitation, and water systems in the Indus Valley civilisations; decadent pyramids and monuments of Egypt; *Qanats* in Persia and Mesopotamia; Stonehenge in UK and the Great Wall of China; where functionality and aesthetics paralleled each other. *This is discussed further, with detail in the next Unit.*

Most of the construction efforts were achieved by manual labour and non-mechanised tools, however, two major global occurrences, namely, the Agricultural Revolution and the Industrial Revolution, created a huge demand of infrastructure for, housing, transportation, sanitation, and water management, and eventually for environmental planning and construction management.

1.1.2 Agricultural Revolutions

The **First Agriculture Revolution**, also known as the Neolithic revolution, occurred in 10,000 BC and led to the organisation of modern man as a producer, rather than a hunter-gatherer. While the following millennia focussed on socio-economic development and in turn, creative feats of engineering and design, as discussed above; agriculture continued to be the primary occupation of the common folk. Across all the major civilisations that arose, from those in Indus Valley, Egypt, Mesopotamia, Greece, Rome, Mesoamerica, Europe, Arabia and eventually the new nations and colonies; innovations and developments centred on the methods, tools and techniques to support agriculture, through construction of water supply for irrigation and sanitation, granaries for food storage, tools and processes for production and preservation, etc.

However, by the mid- 17th century, an unprecedented growth in agricultural production from increase in labour and land productivity was observed in two nations, Britain and Netherlands. This was notably the **Second Agriculture Revolution**, and it spread across the other nations in Europe and their colonies in East Asia and North America over the next two centuries until the late 19th century. While the Dutch observed increase in the agricultural output per labourer, the British experienced the abundant outgrowth of output with respect to the population.

The strategic developments and innovations of the British Agriculture Revolution that shaped the era were; introduction of new crops, selective breeding and the method of crop rotation; improved design of the plough called the Dutch swing plough; and institution of land reforms, such as, land conversion, drainage and reclamation, increase in farm size, and enactment of Enclosure Act. Civil engineering played a significant role here, in the development of transportation infrastructure – roads, canals, railways, as well as social infrastructure, such as, development of national markets, supplemented with policies free of tariff, tolls and customs.

It is interesting to note that, while abundant food supply led to rampant increase in population, it eventually led to loss of labour force engaged in agriculture, leading to a need for new occupations. With time, though the inland production was countered with cheap imports of food supply, this revolutionary spike ensured a complacent food reserve that allowed a shift in the National priorities, from agriculture to industry. A new working class, seeking urban employment in the upcoming industries, emerged as a result and fuelled the onset of the Industrial Revolution.

The first half of the 20th century was rife with war and caused major setbacks in the developed countries. It is also at this time that several colonised nations sought their independence, further adding to the need for re-building and self-sufficiency.

The **Third Agriculture Revolution**, popularly known as the ‘**Green Revolution**’, transpired as a response to global hunger and poverty, dilapidated infrastructure, and scarce and contaminated natural resources by mid-20th century. New technologies and cutting-edge research in the areas of high-yield crops, hybridized seeds, genetically modified organisms (GMOs); use of chemical fertilizers and pesticides; methods of controlled irrigation and mechanized cultivation; characterise this era.

MS Swaminathan, agronomist and parliamentarian, lovingly called the ‘*Wheat man of India*’ invited Norman Borlaug, the ‘Father of the Green Revolution’, at the brink of a famine in 1961, and initiated collaboration with the Ford Foundation to import newly developed variants of wheat and later rice. Punjab became the birthplace of the **Indian Green Revolution** or ‘**Harit Kranti**’ and the Government of India facilitated programs to support farmers in the use of agrochemicals and irrigation. Late Prime Minister Shri Lal Bahadur Shastri, gave the slogan ‘*Jai Jawan Jai Kisan*’. Today, India is the world’s largest rice exporter.



Fig.1.2 : ‘*Harit Kranti*’ - The Indian Green Revolution

Presently, several strategies and contributions of the 'Green revolution' is being questioned, despite the fact that it addressed world hunger, poverty, and reconversion of land for agriculture. The decrease in food security and increase in production of export crops, has been unwarranted results of the same. However, while a rehauling of perspectives is underway to ensure long-term sustainability, some of the fundamental innovations and engineering contributions of this time set the tone for the future.

1.1.3 Industrial Revolution

While the North American colony was the primary producer of cotton, India was compelled to cultivate indigo on large-scale, despite several revolts by farmers. These two crops propelled the production of indigo-dyed cotton fabric, which in turn, thrust the need to improve the existing hand production to machines and mills, leading to the evolution of the design of cotton mill - from *Spinning Jenny* (1764) to the *Power Loom* (1787), to meet growing demands and establish a monopoly in international trade. This marks the **First Industrial Revolution**, spanning approximately 100 years from 1760 – 1850, characterised by mechanisation, fuelled by abundant coal supply and the optimisation of the design of steam engine by James Watt in 1765. new profession in engineering – mechanical engineering, was born, and several machine and production line innovations were developed.

Studies and experiments looking into the phenomena of electricity, by Benjamin Franklin, James Watt and Alessandro Volta, eventually gave rise to the **Second Industrial Revolution** between 1850 – 1917 with electrical energy as a new source of power that enabled mass production. However, by the mid-19th century, the working conditions in the factories across Europe worsened and there was worker unrest leading to The Great Reform Movements, thereby slowing the industries. By early 20th century, World wars and demand of independence by colonies, further led to the slowing of Europe as a forerunner in industry, and USA emerged as the promised contender.

The span between 1877-1917, termed as the '**Gilded Age**' of America by satirical author, Mark Twain, is characterised by massive economic growth, fuelled by development in the areas of manufacturing, railroads, telecommunication, automotive, product and industrial design, in the newly united federate, post-civil war. A centralised government and growing bureaucracy looked at expanding their jurisdiction and focussed on public services, through building of extensive rails and roads, telecommunication infrastructure, '**mass production**' industries and support facilities for public health and sanitation. A new connected world gave a global overview to markets, businesses, politics, and the invention of the lightbulb by Edison allowed prolonged working hours, furthering the industrialisation agenda.

The role of engineers, sanitarians and public health scientists grew in importance, while new domains of **product design and industrial design** came into being. Some of the common products that are used till date, came into conception during this era, such as, the foot-powered sewing machine, patented by Isaac Singer in 1851; the electric iron, invented by Henry Seeley in 1882; the first drum washing machine, patented by James King in 1851; and the first

mechanical, hand-cranked, dishwashing device made of wood, was registered in 1850 by Joel Houghton.

Another key area of innovation continued to be **telecommunication**. Graham Bell invented the telephone in 1876 and founded the AT&T company. While Edison invented the phonograph - a sound recorder, in 1877 and the kinetoscope - a peephole video player, in 1892.

Transportation remained a focus and a remarkable innovation that took place at that time was the design of the 'Quadricycle' by Henry Ford in 1896, that featured four bicycle wheels powered by a four-horsepower, internal combustion engine fuelled by petroleum and not steam. This led to the establishment of the Ford Motor Co. in 1903, which became the America's largest car manufacturer, housing a moving assembly line (*refer Fig. 3*) that made the Ford model T, a household name. That same year, the Wright brothers took first flight in the powered airplane, the 1903 Wright Flyer.



Fig. 1.3 : Assembly Line at Ford Motor Co. (source : www.corporate.ford.com)

*The use of **new source of power – electricity and petroleum**; new materials– iron, steel, glass, rubber; new techniques and design of engineered systems – turbines, engines, motors, conveyors, etc.; new technology, such as, Babbage's Analytical Engine or the first computer, led to mass-production, eased transportation and improved computation and telecommunication. Unfortunately, the early 20th century saw the onset of two World Wars, during which, while great scientific and engineering progress was made, catastrophic societal and environmental damage was onslaught.*

1.1.4 Digital Revolution and Industry 4.0

An era of rebuilding began around 1947, which is noted as the Third Industrial Revolution or Digital Revolution, characterised by the adoption of digital technologies, many of which had its precursors developed during the World Wars. It is also at this crucial time in history when India received her independence and was faced with many challenges, and took time to catch up to the state of the art.

The early years of the '*Information Age*' saw several firsts. In 1943, the ENIAC (Electronic Numerical Integrator And Computer) was developed; In 1947, the first working transistor was designed at Bell Labs, and later the MOSFET or MOS transistor was developed. In parallel, at Fairchild Semiconductor; the first monolithic integrated circuit chip was developed in 1959, alongside research and development in the area to improve the MOS chips, paving the way for the first microprocessor, Intel 4004. This technology also led to the development of image sensors for future digital cameras. In the 1960s and 70s, early CAD (Computer aided Drawing) software were sprouting, with Sutherland's SKETCHPAD, popularly nicknamed, 'Robot Draftsman' and Hanratty's Automated Drafting And Machining (ADAM).

1969 saw the breakthrough invention of the 'internet', when a message over the first wide-area network, ARPANET, was sent to public. This further led to development of inter-networking protocols, and with the introduction of the first home computer in 1970, a proliferation of digital technology and information sharing became the new way of life. Apart from utilitarian needs of digital record keeping, computation and automation, computers also led to a thriving video gaming industry that excelled in the development of first game consoles, game graphics and arcade games. Over the next decade, industrial robots, CGI in film and animation, electronic music and signages, were widely incorporated and eventually, the first mobile phone by Motorola in 1983 and the first digital camera in 1988 came into being.

The internet arrived in India in 1989, through the development of our indigenous network commissioned by the Department of Electronics (DoE), modelled on the ARPANET, named ERNET (Education and Research Network) and as a start, it connected IISc Bangalore, 5 IITs, NCSDCT (National Centre for Software Development and Computing Techniques) and DoE. Hardware and computers were already being imported by 1981 and India was already exporting software. In 1984, C-DoT (Centre for Development of Telematics) was established under the leadership of Sam Pitroda to design digital exchanges, but later expanded to develop software applications. In 1993, satellite link became operational in Bangalore Software Technology Park, and by 1998, VSNL launched Internet services that enabled companies operating from these parks to engage international clients. The Information Technology India market accounts for 9.3% of India's GDP and 56% of the global outsourcing market (IBEF, 2022) and all six Indian brands feature among the top 10 fastest-growing IT Services brands over the course of 2020-2022.

While digitisation remains integral to our present day lives, a new technological advancement is on the horizon, termed as **Industry 4.0 or the 4th Industrial Revolution**,

characterised by artificial intelligence, advanced robotics, internet of things (IoT), cloud computing and cyber-physical systems (CPS). ‘i4.0’, coined in 2015, leverages interconnectivity and big data to facilitate quick and decentralised decision-making and allows to cater to customised solutions for individual needs, in contrast to ‘mass -produced’ solutions, and imbibes increased automation, improved communication, Machine-to-machine as well as Human-Computer Interactions, and self-monitoring and autonomy.

Several i4.0 technologies, such as, Building Information Modelling (BIM) with cloud technologies, additive manufacturing or 3D printing, AR/VR (Augmented and Virtual Reality), drones and unmanned aerial vehicles, etc. is widely being used in the Architectural design, Engineering and Construction (AEC) domain today.

1.2 MAJOR BREAKTHROUGHS AND INNOVATIONS

From early man’s tools to present technology of using drones, unmanned vehicles, and cloud computing; the major breakthroughs and innovations that moulded what civil engineering is today is as follows:

Tools, the first breakthrough that enabled construction, were developed from available natural materials, dating back to the Neolithic age. Stone axe with wooden handle, bone hammer, stone adze, celt, flake tools, sickle, drills and even, sledge, have been discovered from that era. Ancient tenements and structures were primarily built of raw natural resources that were abundant in the region, such as, rocks and stones, timber and bamboo, mud, and clay, etc.

Bricks, were a major innovation and till date, is the integral building-block of most structures. Sun-baked mud bricks or ‘*adobe*’, cemented by lime mortar, was extensively used in Indus Valley and Egypt. Later, kiln-fired, and glazed bricks were developed in Mesopotamia. Large rocks were also chiselled and faced into massive bricks. This in turn, led to innovations in new tools, such as, ramp, lever, lathe; and techniques, such as, butterfly interlocking, method of drilling stone, enamelling; led to development of complex and tall structures, such as, pitched-brick vaults, and pyramids.

Cranes, pulleys, and jibs to raise construction materials to great heights, employing metal cramps to join large stone blocks, and development of early ‘construction drawings’, to build mega-structures, such as, groin vaults, arch bridges and the early multi-storied buildings were some of the contributions of the Greek.

Road design and construction, as we know it today, was first formidably developed by Romans and the first known roadway, the **Appian Way** or the ‘*Queen of the roads*’, was constructed in 312BCE, connecting Rome with its allies in Capua. The design comprised of digging shallow trenches with retaining walls on either side, filled with layers of; levelled earth and mortar or sand topped with rocks, crushed rocks or gravel cemented with mortar and surfaced with blocks of cut rocks, arranged pebbles, iron ore or hardened lava.

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Roman cement, developed by adding volcanic ash, called *pozzolana*, in lime mortar to make it harden under water, is one of the major breakthroughs in the history of civil engineering and construction. This further led to the development of Roman concrete, where concrete made of rubble and mortar was filled inside stone or brick formwork, which was later replaced with the use of removable wooden shuttering; this technique allowed building of arches, barrel vaults and domes over large spans.

Romans also introduced, the use of lead or '*plumbum*' for pipes and roof covering, thus being the origin of the word plumbing; and centralised heating achieved through raising the floor above a wood or coal fire exhaust. They also made use of glass for windows and in mosaics.

Iron reinforcement dates back to the time of the ancient times when cramps and bolts were in use to either hold together stone blocks or timber members in trusses. Later in the 15th century, the same were used by Brunelleschi in the design of the dome, where two layers of domes met at the top in an open stone compression ring and iron cramps held together tie rings running horizontally between ribs. The design was improved in cupola of St. Peter's Basilica, where three continuous iron chains held the dome in tension.

Pile driver, a tool to drive 'piles' or vertical, pole-like, structural members of deep foundation, driven deep into the ground where the soil is loose, to support piers, bridges, cofferdams, etc., was invented in 1500.

First railway line has been referred to in 1515, having wooden rails and a hemp haulage rope through a treadwheel operated by manual power. Several funiculars have been spotted across Europe, however, the world's oldest operational railway - the Middleton Railway, was built in 1758 in Leeds, UK.

Drafting and surveying tools like, the line gauge, plumb line, the carpenter's square, the spirit level, and the drafting compass were developed in the 17th century.

Structural iron was introduced as a building material with the availability of iron, soon replacing wood and charcoal. Iron ore was smelt with the use of coke on a mass scale in 1702. Christopher Wren used iron hangers to suspend floor beams at Hampton Court Palace, and iron rods to repair Salisbury Cathedral and strengthen the dome of St Paul's Cathedral, and iron columns in the House of Commons. Cast-iron was used for the bridge at Coalbrookdale in 1776, while wrought iron was used for the roof structure of the Louvre, Paris. Later in the 19th century, the first two exhibitions of The **Great Exhibition of the Works of Industry of All Nations**, or the *Exposition Universelle*, namely, the Crystal Palace in Hyde Park, London, and the Eiffel Tower, Paris, were also exemplars of iron structures.

Modern road design developed in the 18th century, with method improvements and new technique proposals by Tresaguet, Metcalf, Telford, and McAdam. The present design is an adoption, named the '*Tar McAdam Road*', patented in 1901.



Fig. 1.4 : Ancient building materials and techniques - (Top Left) Egyptian Hieroglyph showing building construction with brick; (Top Middle) Roman Cement, (Top Right) Lead 'plumbum' Pipes, (Bottom Left) Sun-dried Mud bricks of Mesopotamia ; (Bottom Right) Roman Road

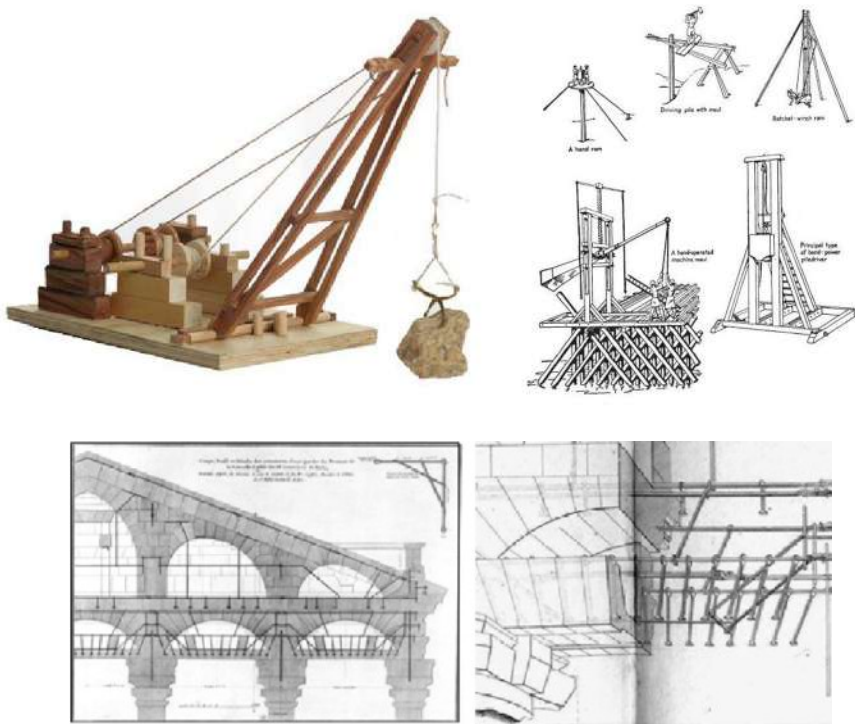


Fig. 1.5 : Ancient construction tools and techniques - (Top Left) Greek crane, (Top Right) Designs of Pile Driver, (Bottom) Rondelet's drawing of reinforcement iron bars (Blasi and Iori, 2008)

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First iron chain suspension bridge in the Western world was the Jacob's Creek Bridge (1801), Westmoreland County, Pennsylvania, designed by inventor James. However, Union Bridge (1820) is the oldest operational chain suspension bridge.

The **first wire-cable suspension bridge** was the Spider Bridge at Falls of Schuylkill (1816),

Portland cement was patented in 1824 by Joseph Aspdin and its modern common version was developed by his son, William, in the 1840's.

The **first** advertised **prefabricated home** was the "Manning Portable Cottage" conceived in 1830 by London carpenter H. John Manning. Shortly after in 1845, Isambard Brunel designed the prefabricated wood and canvas, Renkioi Hospital, for assemblage on site at the Crimean War.

The **first systematic national building standard** was established with the London Building Act of 1844, and by 1855, Metropolitan Board of Works was set up. Meanwhile in the USA, the City of Baltimore passed its first building code in 1891 and by 1908 a formal building code was rafted and adopted.

Reinforced concrete was invented in 1849 by Joseph Monier and the first reinforced concrete bridge was built by the inventor-engineer.

Building of the **Panama Canal** between **1905-14**, was a feat of civil engineering as it was the biggest earth dam in the world and created the largest man-made lake, by deepening of the Pacific and Atlantic **canal** entrances, widening and deepening the Gatun Lake navigational channel. It remains one of the busiest shipping lanes in the world.

Mass production of Steel, was enabled by the Bessemer process, introduced in 1855. However, the move to steel as an important material date back to 1740, with the development of the crucible steel technique by English inventor, Benjamin Huntsman, who also established a steelworks at Sheffield, England. This was further supported with the use of steam engines to operate heavy machinery and the invention of steel roller for steel production by Henry Cort in 1783.

Along with **mass-production of Glass** and the urge to explore non-traditional design and construction in mid- 19th century, **the first skyscrapers**, namely, the Empire State Building, Rockefeller Center and Chrysler Building, in New York were conceived. The Empire State Building was constructed in only 13 months, towering to a height of 443 meters and 102 stories, it was the tallest building of its time in 1931.

Prestressed concrete invented by Freyssinet in 1928 and further applied it to develop precast segmental construction.



Fig. 1.6 : Panama Canal (*Top*) A Cross Section Canal, (*Bottom*) Picture

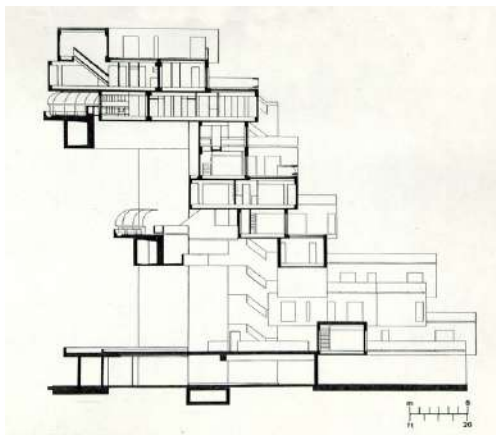


Fig. 1.7 : Modular construction - Habitat 67, Montreal (*Left*) Section drawing, (*Right*) Photograph

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The following innovations are of particular importance in modern construction and will be further discussed in later Units.

Modular construction of home systems first materialised in 1933 with the Winslow Ames House by Robert W. McLaughlin and his firm, American House Inc. Cemesto, a panel board made from sugarcane, patented by the John B. Pierce Foundation, was used and in 1942 the US government employed Skidmore, Owings, & Merrill (SOM), to come up with a scheme called “Flexible Space”. One of the most remarkable prefabricated, modular megastructures remains the Habitat 67, by Moshe Safdie for the Expo in Montreal. The market is projected to increase from \$92.18 billion in 2018 to \$130 billion in 2030.

Development of CAD (Computer Aided Drawing) software can be traced back to the development of PRONTO, the first commercial numerical-control programming system by Dr. Patrick Hanratty in 1957, and to the first ‘Robot Draftsman’ or SKETCHPAD application developed by Ivan Sutherland’s during his doctoral work at MIT in 1963, that used a GUIA graphical user interface that facilitated human-computer interaction through visual aids or icons.

BIM (Building Information Modeling) as a concept can be traced between 1970-1990 to the first software tools developed for modelling buildings, i.e., the Building Description System, GLIDE, RUCAPS, Sonata, Reflex and Gable 4D Series. The term ‘Building Information Model’ first appeared in a 1992 paper authored by van Nederveen and Tolman, but it was only in 2003 that Jerry Laiserin, acknowledged the contributions of Autodesk, Graphisoft and Bentley systems and standardized the term as a common name for the digital representation of the building process.

Construction 3D Printing is the technology of using additive manufacturing technique through computer-controlled activities of; sequential extrusion of material, such as, 3D concrete, powder bonding and additive welding, through Autonomous Robotic Construction systems (ARCs) or Freeform Construction. While empirical development of this technology started out in early 2000s, it may be traced back to the 1950’s when robotic bricklaying and on-site automated fabrication was being explored. Presently various applications, such as, fabrication of houses and construction components, building bridges and canals, and even artificial reefs are being made (*Refer KNOW MORE Section to know about India’s Research in this area*)

Smart buildings and Building Automation systems are terms used in conjunction today to refer to buildings that use IoT technology to monitor and connect various devices, sensors, hardware, software to manage various building services, such as, HVAC, lighting, fire protection, security, and access control, etc.

While various innovations have transformed design, engineering and construction, good old-fashioned soundness of the principles of civil engineering remains fundamental to the success of a built-environment and the hallmark of a good engineer. This is elucidated by the infamy of the Tower of Pisa, popularly called the ‘**Leaning Tower**’, near Florence Italy, and the remarkable team that saved it from collapsing. The 56m freestanding *campanile*, or bell tower, began to lean within five years of work

commencing in 1173, caused by shallow foundations in unstable sub-soil. It took 200 years to complete, all the while continuing to tilt, until 1993 when it was reportedly at an alarming 5degree tilt. This lean was corrected by a team of experts that devised a solution called ‘soil extraction’, where a series of tunnels on the side opposite to the tilt was dug to drain the soil and remove small amounts of soil, while reinforcing the foundations with 15m concrete pillars and harnessing with steel cables to pull it back in position, till it self-corrected by 2001.

Every breakthrough and innovation, however, has profound consequence associated with it. These impacts maybe positive, like development of civic amenities, transport facilities, telecommunication, and global trade; or negatives, like deforestation, loss of indigenous flora and fauna, colonisation, poor labour conditions and poverty; and are increasing at a global level with the advent of globalisation and enhanced connectivity. Civic life is nested on Society, Economy, and Ecology (or Environment), known as the three pillars of Sustainability, with the overarching intent to improve quality of life for today’s populace and future generations.

1.3 SUSTAINABLE DEVELOPMENT: PRESENT DAY WORLD AND FUTURE PROJECTIONS

With Industry 5.0 on the horizon, a long-term, sustainable point of view for development is the need of the hour. The concept of ‘Sustainability’ was first introduced in the context of development in the book ‘*Our Common Future*’, popularly known as the Brundtland Report, from the World Commission on Environment and Development (WCED), published in 1987 by the United Nations. It defined the term as “*meeting the needs of the present without compromising the ability of future generations to meet their own needs*” and propagated the principles of ‘Sustainable development’ that focuses on environmental protection and social equality along with economic growth. This eventually laid the foundation for formulating global Sustainable Development Goals (SDGs) in 2015, adopted by 193 committed member states of the United Nations with the projection of meeting the goals by 2030.

1.3.1 The steady erosion of Sustainability and Mitigative Actions

Following the world wars and the advent of the Third Agricultural revolution, two evocative pieces, namely, ‘*The Silent Springs*’ (Carson, 1962) which argued against excessive use of pesticides like DDT and its harmful impact on several species, and ‘*Scarcity and Growth*’ (Barnett and Morse, 1963) which empirically established the burden on natural resources, can be noted as the beginning to a paradigm shift at a global level to address issues of sustainability at the intersection of social and environmental or natural ecosystems. However, the impending ecological crisis was first noted in the previous century, with Swedish scientist Arrhenius, in 1896, predicting the change in surface temperature owing to greenhouse effect caused due to increased fossil fuel use, later corroborated by Guy Callendar in 1938 who connected carbon dioxide increase in earth’s atmosphere to global warming. Research across 1940’s through the 1960’s revealed the implications of CO₂ emissions; with Plass reporting that CO₂ in atmosphere captures infrared radiation, otherwise lost to space, and Keeling producing concentration curves

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for atmospheric CO₂ showing a downward trend in annual temperatures. The 60's was a decade of ecological strife, with Monaco opposing French plan to dump radioactive waste in the Mediterranean Sea, and catastrophes such as, the Torrey Canyon oil tanker spillage off the coast of England and the Santa Barbara oil spill. These findings and events led to the first Earth Day celebration in 1970 and the creation of the US Environmental Protection Agency (EPA). Further, an inter-governmental conference of experts for Rational use and conservation of Biosphere (UNESCO, 1970) was convened, and in 1990 **Earth Day** went global.

India, too, saw its share of conservation movements. The *Chipko Movement* of 1973 was a non-violent social and ecological '*andolan*' by rural villagers, mostly women, protesting government backed logging in the Himalayan regions of Uttarakhand, India. Inspired by the Bishnois of Khejari, who were killed trying to protect their sacred trees in 1730's in the kingdom of Marwar, the Chipko protestors stood hugging trees and in 1987, this movement was recognised with the Right Livelihood Award for its conservational efforts. This further inspired several other similar conservation movements across Rajasthan, Bihar, Himachal Pradesh, the '*Appiko movement*' in Karnataka and the Western Ghats in 1983, as well as campaigns to protect the canopies planted along the Grand Trunk Jessore Road in West Bengal, as recent as 2017. India enacted the *Wildlife (protection) Act* in 1972, *Water (Prevention and Control of Pollution) Act* in 1974, *Air (Prevention and Control of Pollution) Act* in 1981, followed by the *Environment (Protection) Act* in 1986. The 21st century has seen several amendments and additions, such as, special regulations to Environment Act for ozone-depleting substances in 2000 and Coastal zone notification in 2018, The *Energy Conservation Act* 2001, and *Biological Diversity Act* 2002. While most of these aimed at protection and conservation of natural Environment, in 2006 the enactment of the Forest Rights Act (FRA) for scheduled tribes and other traditional forest dwellers, recognised the interdependency within the socio-ecological ecosystems and introduced an unprecedented reform at the intersection of the Ecology and Social dimensions of sustainable development.

While development remained the key motivation, various strategies, policies and acts were established between 1967- 1987 to further the sustainability discourse in the West. The *Clean Air or Air Quality Act* (1967), the *National Environmental Policy Act* (PEA, P.C.E.A., 1969) and the *Endangered Species Act* (1973) were passed by the US Congress; and legal action for environmental damages was pursued by Environment Defense Fund (EDF 1967, www.edf.org). Several books and papers, such as, '*Only One Earth*' (Dubros and Ward, 1971), '*Limits to Growth*' (Meadows et al., 1972), '*Polluter Pays Principle*' (OECD, 1972) , '*World Conservation Strategies*' (Talbot, 1980, released by IUCN), '*Global 2000 Report*' (Barney, 1980 and Council on Environmental Quality), and UN's '*World Charter for Nature*' (un.org) were published; and various events, such as, '**Habitat**' - the UN Conference on Human Settlements in 1967, the International Conference on Environment and Economics by OECD in 1984, and the UN General Assembly of 1987, where the foreword of '*Our Common Future*' was presented, took place. This culminated into the '*Agenda 21 Declaration of Environment and Development*' (UNCED, 1992), an action plan proposed for the 21st century in the 1992 Rio Earth Summit. The UN Conference on Sustainable Development or **Rio+20** was held twenty years later in 2012, where the process to develop measurable goals and targets as a set of Sustainable Development Goals (SDGs) was adopted, along with green economy policies, and in 2015 the fifteen-year plan to '*2030 Agenda for Sustainable Development*', bearing the

17 goals and 169 targets were committed to by 193 member nations. Currently, between October and December 2022, the Rio +30 conference is being organised.

1.3.2 Sustainable Development Goals and Global Impact

The present condition, post pandemic, with respect to each SDG (sdgs.un.org) is as follows:

1. **No Poverty** – COVID 19 pushed 8 million workers into poverty worldwide and working poverty rate climbed for the first time in two decades. Furthered by inflation, wars and political crisis, and disaster there is an ongoing migration and refugee crisis making ending poverty the foremost priority.
2. **Zero Hunger** – 1 in 10 people worldwide are suffering from hunger and 1 in 3 people lack regular access to adequate, in addition, food shortages and soaring food prices affected 47% of countries in 2020, up from 16% in 2019. Ending hunger, achieving food security and improved nutrition, and promoting sustainable agriculture are some of the primary targets.
3. **Good health and well-being** – The COVID 19 led to 15 million deaths and essential health services in 92% of countries got disrupted (2021), leading to reduction in global life expectancy and prevalence of anxiety and depression. Thus, ensuring healthy lives and holistic well-being at all ages is a critical goal.
4. **Quality Education** - Between 2020-21, over half of in-person instruction at schools were missed by 147 million children, and it is estimated that 24 million learners worldwide, from pre-primary to university level, may never return to school. At the same time, many countries have started improving classroom infrastructure as 25% primary schools still lack electricity, drinking water and basic sanitation, and around 50% lack computers and internet access. In addition, skill and competence development is a driver for industry readiness, improved economy and social upliftment, making quality education a major goal.
5. **Gender Equality** - While there was a rise in women employment in 2019, the 45% global employment loss in 2020 set back the equal representation pace. It was also revealed through a 15 year survey, that 1 in every 4 women, accounting to 641 million women, were subject to violence at least once in their lifetime. Therefore, exalting women is decisive towards a better and inclusive future generation.
6. **Clean water and sanitation** – World's water ecosystems are degrading at alarming rates, with over 85% of wetlands being lost, while 733+ million people live in countries with high and critical levels of water stress. This goal aims to ensure availability and sustainable management of water and sanitation for all, as 4 times increase in requirement is estimated by 2030.
7. **Affordable and clean energy** - While progress in energy efficiency is underway, and total renewable energy consumption has increased, the annual energy-intensity rate needs to go up from 1.9% presently to 3.2% by 2030. The key hurdles are; slowdown in electrification due to the challenge of reach, the use of inefficient and polluting cooking systems, and decline in financial flows to develop countries for renewables, making affordable and clean energy a priority.
8. **Decent work and Economic growth** – While global unemployment plummeted, child labour worldwide continues to account to 1 in 10 children. Presently annual growth rate of global real GDP per capita got affected by rising inflation, supply-chain disruption, policy uncertainties,

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etc. Thus, full, and productive employment is crucial for decent work and sustainable economic growth.

9. **Industry, innovation and infrastructure** – While global manufacturing has bounced back and is on a steady rise, over the COVID crisis it was noted that high-tech industries are more resilient than lower-tech counterparts, with small scale industries lacking financial support and loss in manufacturing jobs. Therefore, building resilient infrastructure, promote inclusive and sustainable industrialisation and foster innovation is key.
10. **Reduced inequalities** - The pandemic caused a first rise in between-country income inequality, and global refugee figures hit a record high. In addition, discrimination on at least one of the grounds prohibited under international Human Rights Law is still faced by 1 in 5 people, making reducing inequality a major global goal.
11. **Sustainable Cities and Communities** - As rapid urbanisation occurs, issues of; polluted air breathed in by 99% of urban population, municipal solid waste of which only 55% is managed, and 1 billion slum dwellers indicate the need to make cities and human settlements inclusive, resilient, and sustainable.
12. **Responsible Consumption and production** – With reliance on natural resources increasing over 65% globally from 2000-2019 and large amounts of food is either lost between harvesting and reaching markets, or wasted at the consumer level, making sustainable consumption a pivotal goal.
13. **Climate Action** – Droughts estimated to displace 700 million, extreme weather, and other natural disasters estimated to increase by 40% by 2030, climate catastrophes are surmounting. 70-90% of coral reefs have diminished and CO₂ emissions have gone up to its highest in 2021, while climate finance has dropped with a shortfall of 100 billion dollars in 2019; showcasing the grim state of affairs and the urgency for climate action as a global goal.
14. **Life below Water** - Oceans are our planet's largest ecosystem and is endangered due to ocean warming, eutrophication, acidification, over-fishing, and plastic pollution; with over 17+ million metric tons of plastics choking the ocean and acidification hindering the ocean's capability to absorb CO₂ emissions to moderate climate change. Conservation and sustainable use of marine resources is a necessity.
15. **Life on Land** – Almost 90% of global deforestation is due to cropland expansion and livestock grazing, and around 40,000 species are at the risk of extinction. Protecting, restoring, and promoting sustainable use of terrestrial ecosystems, managing forests sustainably, combating desertification and address land degradation and biodiversity is our moral duty.
16. **Peace, Justice and strong Institutions** – With a quarter of the world's population living in conflict-affected countries and corruption and bribery rampant, providing justice for all, and building effective, accountable, and inclusive institutions at all levels is imperative for peaceful societies.
17. **Partnerships for the Goals** - Post pandemic, with rising debt burdens threatening developing countries and global Official Development Assistance (ODA) declining for SDG data, it is vital to strengthen the means of implementation and revitalise the global partnerships for sustainable development.

Each SDG has a number of **Targets**, each measurable by indicators; for example, Target 1.1 '*By 2030, eradicate extreme poverty for all people everywhere, currently measured as people living on less than \$1.25 per day*' can be measured through the **indicator** 1.1.1 '*Proportion of population living below the international poverty line by sex, age, employment status and*

geographical location (urban/rural)'. India, as the fifth largest economy and the second largest in population in the world, plays a crucial role in the achievement of Agenda 2030 and the future. The **SDG India Index**, an aggregate measure presented in a comprehensible, interactive dashboard (2020-21; <http://sdgindiaindex.niti.gov.in>), for 16 goals with 115 quantitative indicators and a qualitative assessment of SDG 17, tracks and monitors progress of all states and union territories.



Fig. 1.8 : Sustainable Development Goals (source : www.sdg.un.org)

As the world moves towards sustainable development, the role of civil engineering becomes pivotal, needing to shoulder the responsibilities of designing and developing sustainable solutions.

1.4 EVALUATING FUTURE REQUIREMENTS

Brundtland's Commission (1987), in its definition of Sustainable Development, stressed on the "*future generations*" and the ability to "*meet their needs*". These needs and whether or not the ability to attain them by future generations can be assessed by sustainability indicators attributed to each SDG and its respective targets and are measured with various monitoring applications that capture quantitative data, further collated and represented in Indices to offer a holistic view on sustainable development.

1.4.1 Sustainability Indicators

Sustainability indicators are measurable aspects of the three dimensions of sustainability – social, environmental, and economic, and are essential for monitoring and calibrating the performance and quality of the sustainability goals. In addition, these help in decision-making by providing aggregated information to incorporate physical and social science into actionable items and help in predicting early warnings to prevent setbacks.

Division of Sustainable Development (DSD) and Statistics, under the UN Department of Economic and Social Affairs drafted the first set of indicators, which was later collated with

methodology sheets for each indicator into a single publication known as the ‘blue book’ (1996). The same was revised in 2001 having 58 indicators, classified into themes, such as, poverty, governance, health, education, demographics, natural hazards, biodiversity, consumption, and production patterns, etc. Every theme, in turn was categorised into sub themes with core indicators, for example – Poverty is sub-thematised into; ‘*income poverty*’ measured by the indicator ‘proportion of population living below national poverty line’, and ‘*income equality*’ measured by the indicator ‘ratio of share of national income of highest to lowest quintile’, etc. Certain sub-themes maybe measured by several indicators, such as, ‘*Education level*’ is indicated by: ‘gross intake ratio to last grade of primary education’, ‘net enrolment rate in primary education’, ‘adult secondary (tertiary) schooling attainment level’. Interestingly, several indicators have links to more than one theme, such as, ‘proportion of population with access to safe drinking water’ is primarily applicable to Poverty and Health, but also has secondary linkages to governing water utilities, and in turn, to Governance theme.

1.4.2 Monitoring: Methodology and applications

Monitoring is a major task for assessing the implementation and impact of a strategy for sustainable development. A variety of data and statistics are required for monitoring and to capture this, methodologies to measure the indicator, collect the data accurately and timely, outline the apt unit of measure and method of computation, etc are developed by three key bodies, namely the UNSC, HGL-PCCB, and IAEG-SDGs. The latter has selected custodian agencies, such as, UNEP, WHO, World Bank, FAO, UNESCO, OECD, etc, for periodically collecting and updating data on indicators, across themes, at global, regional, and national levels. **Ten principles for Global monitoring indicators** (UNSDSN, 2015) entail;

1. Limited in number and globally harmonised,
2. Simple, single-variable indicators, with straightforward policy implications,
3. Allow for high frequency monitoring,
4. Consensus-based,
5. Consensus-based, in line with international standards and system-based information,
6. Constructed from well-established data sources,
7. Disaggregated (by sex, age, location, income, spatial - rural/urban)
8. Universal,
9. Mainly outcome-focused,
10. Science-based and forward-looking,
11. A proxy for broader issues or conditions.

Indicators to be measured are categorised with respect to global, national and thematic, and Data in turn, is sourced from census data and *Household surveys* (Demographic and health surveys, Fertility and Family Surveys, Reproductive health surveys, Labor Force survey, R&D Surveys etc); *Administrative data* (formal waste collection and management data from municipalities, National production, international trade statistics); Civil registrations

(birth certificates, death certificates, etc), *Vital statistics* and *health records*; *School-based or citizen-led learning assessments*; as well as from *Geographic Information Systems* (GIS) that uses remote sensing, GPS (global positioning system), aerial photographs, LiDAR (Light Detection and Ranging) and satellite imagery from US Geological Survey/NASA Landsat data, for mapping and surveying.

GIS is an application that uses geographical and spatial data in conjunction with attribute (additional information in tabular form) data to map, analyse and assess indicators. It uses imagery data-type, which includes aerial photos, satellite images, thermal images digital elevation models, scanned maps, land maps, land classification maps, etc. Satellites are used to scan the earth, either through - scanning mirrors that scan quickly, back and forth over an area while taking an image, as used by Landsat or Multispectral scanners, called ‘*whiskbroom scanning*’, or via ‘*push broom*’ scanning where a row of silicon detectors take images as the satellite flies over an area. GIS not only aids in cartography but helps geospatial mapping and analysis of real-world problems to offer insights through visualisations. The ‘*Global Geodetic Reference Frame*’ (UNDESA) allows the precise determination of locations and quantification of changes, useful for various indicators through mapping physical infrastructure, buildings and settlements, population distribution, transport networks, elevation and depths, water and land cover and use, etc.

1.4.3 Human Development Index and Ecological Footprint

While SDGs are a tool for addressing developmental progress, there are other indices that also help measure the impact of human development.

One such, is the **Human Development Index (HDI)** that offers an “*alternative, single number measure of capturing progress in three basic dimensions of human development: Health (life expectancy at birth), Education (expected and mean years of schooling) and Living standards (Gross National Income per capita)*” as defined by the Human Development Report (UNDP). This index was further refined and reintroduced in 2010 as ‘*Inequality adjusted HDI*’ (IHDI) accounting for the inequalities between the various nations, noting that HDI maybe perceived as a potential while IHDI reflects the reality. Another important indicator is the **Ecological Footprint (EF)** that measures the human - a person or a group, demand on natural capital. It estimates the productive land and sea area needed to support a population in terms of its consumption of renewable resources and absorption of waste generation. It shows whether a country is living within the biocapacity of its own territory or is drawing on the ecological capital of other nations. Both HDI and EF have attracted criticism owing to aggregation as a single number upon summation of different indicators, each having different units, and in turn, missing the finer details that represents the scenario realistically.

The upcoming decades, bound by a common developmental discourse since the industrial revolution, and further today, by digitisation and globalisation, faces the inadvertent need to strive towards Sustainable development and to understand the importance of innovation and social responsibility, particularly through Civil engineering in shaping and impacting the world.

UNIT SUMMARY

The Unit covers, in brief, the developmental history from ancient to pre-industrial era, giving an overview of the pockets of civilisations across the world and their prowess in engineering, architectural and construction capability. The Unit further traces the oncoming of the second agriculture revolution and how it prompted the first industrial revolution, followed by decisive events and occurrences termed as further revolutions – green revolution, digital revolution, etc., and culminates the discussion with the impact of the above on the natural environment and social welfare, leading to the importance and need for sustainable development, and how present measures have been instituted to achieve the same.

EXERCISES

I. Multiple Choice Questions

Q. 1.1 What is the system of transporting water from aquifers and wells to the surface through underground aqueducts, found in Persia and Mesopotamia?

- (a) Baoli
- (b) Qanat
- (c) Reservoir
- (d) Hammam

Q. 1.2 When did the 'Green Revolution' take place?

- (a) late 19th century
- (b) early 20th century
- (c) 10,000 BC
- (d) mid 20th century

Q. 1.3 Who coined the term "Gilded Age"?

- (a) Mark Twain
- (b) FD Roosevelt
- (c) Henry Ford
- (d) Earnest Hemingway

Q. 1.4 Under who's leadership was C-DOT (Centre for Development of Telematics) established in India?

- (a) JRD Tata
- (b) Manmohan Singh
- (c) Sam Pitroda
- (d) Sundar Pichai

Q. 1.5 In which year was the '2030 Agenda for Sustainable Development' adopted?

- (a) 2020
- (b) 2015
- (c) 1987
- (d) 2000

Answers of Multiple Choice Questions: 1.1 (b), 1.2 (d), 1.3 (a), 1.4 (c), 1.5 (b)

II. Short and Long Answer Type Questions

Q. 1.6 What is Sustainable Development? What are the SDGs?

Q. 1.7 What are the different modes of monitoring SDGs? Enlist the Ten principles of Global monitoring indicators as per UNSDSN (2015)?

Q. 1.8 Define Human Development Index (HDI) and Ecological Footprint? What are the shortfalls of these indices?

KNOW MORE

Civil Engineering during the Industrial Revolution in Britain



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India's IT Revolution



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Sustainable Development Goals and Indicators



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